6R 1100/1300/1500
OM 470/471/473

Operating Instructions

optimized by

mtu

manufactured by

Symbols

WARNING

Warning notes make you aware of dangers which could pose a threat to your health or life, or to the health and life of others.

Environmental note

Environmental notes provide you with information on environmentally aware actions or disposal.

! The purpose of material damage warnings is to draw your attention to risks which could lead to damage to your engine system.

i These symbols indicate useful instructions or further information that could be helpful to you.

▶	This symbol designates an instruction you must follow.
▶	Several consecutive symbols indicate an instruction with several steps.
(▷ page)	This symbol tells you where you can find further information on a topic.
▷ ▷	This symbol indicates a warning or an instruction that is continued on the next page.

Welcome!

Familiarise yourself with your engine system and read the Operating Instructions before you use it. This will help you to avoid endangering yourself or others.

The standard equipment and product description of your engine system may vary, depending on individual specifications. This is described on the data card.

The engine systems are constantly updated to be state of the art.

Descriptions may therefore differ from your engine system in individual cases.

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Engine system

In these Operating Instructions, the term "engine system" refers to the combination of the engine and exhaust gas aftertreatment unit. The engine and the exhaust gas aftertreatment system are parts that require certification. The engine system is certified as one unit with the engine and exhaust gas aftertreatment unit assemblies as well as the AdBlue® supply system.

The engine system is certified for the following emissions level:

- EU Stage V

Protection of the environment

Environmental note

Daimler AG has a declared policy of comprehensive environmental protection.

The objective is to use natural resources sparingly and in a manner that takes the requirements of both nature and humanity into account.

You too can help to protect the environment by operating your vehicle/device in an environmentally responsible manner.

Information and notes on driving in an environmentally responsible and fuel-saving manner can be found in the "Operating notes" section (▷ page 29).

Assembly equipment

These Operating Instructions describe all models and all standard and optional equipment available for your engine system at the time of publication of the Operating Instructions. Country-specific deviations are possible. Note that your engine system may not be fitted with all features described. This also applies to safety-relevant systems and functions. Therefore, the equipment on your engine system may differ from certain descriptions and illustrations.

All of the components in your engine system are listed in the data card of your engine system. Data card (▷ page 55).

If you have questions relating to vehicle equipment or operation, please consult any MTU or MTU-authorised Mercedes-Benz Service Centre (▷ page 7).

Genuine Mercedes-Benz parts

Environmental note

Daimler AG also supplies reconditioned assemblies and parts which are of the same quality as new parts. For these, the same warranty applies as for new parts.

If you use parts which have not been approved by Mercedes-Benz, the operational safety of the engine system may be jeopardised. This could lead to malfunctions in safety-relevant systems. Use only genuine Mercedes-Benz parts or parts of equal quality. Only use parts that have been approved for your engine type.

Mercedes-Benz checks genuine Mercedes-Benz parts for:

- reliability
- safety
- suitability

Despite ongoing market research, Mercedes-Benz is unable to assess other parts. Mercedes-Benz therefore accepts no responsibility for the use of such parts in Mercedes-Benz vehicles, even if they have been officially approved or independently approved by a testing centre.

In Germany, certain parts are only officially approved for installation or modification if they comply with legal requirements. This also applies to some other countries. All genuine Mercedes-Benz parts meet the approval requirements. The use of non-approved parts may invalidate the vehicle's general operating permit.

This is the case if:

- they result in a change to the vehicle/device type from that for which the general operating permit was granted
- they pose a possible risk for road users
- they adversely affect the emission or noise levels

Always specify the assembly system ID number (AGS ID), the engine number and the number of the exhaust gas aftertreatment unit when ordering genuine Mercedes-Benz parts. You can also find these numbers on the data card (▷ page 55).

Modifying the engine output

! Increased power could:

- change emission levels
- cause malfunctions
- lead to consequential damage

The operating safety of the engine cannot be guaranteed in all situations.

Any tampering with the engine management system in order to increase the engine power output will lead to a loss of warranty entitlements.

Safety/emergency running program

The engine is equipped with an electronic engine management system that monitors not only the engine and the exhaust gas aftertreatment unit but also itself (self-diagnosis).

If the electronic control system detects a malfunction, one of the following measures is automatically implemented after an appraisal of the malfunction:

- faults during operation are indicated by the corresponding warning lamp (▷ page 23).
- in conjunction with the electronic engine management system, fault codes with additional information can be shown on a display.
- the system switches to a suitable backup function for the continued, albeit restricted, operation of the engine. This includes, for example, torque and speed limitation as well as constant emergency running speed.

Correct use

The engine system may only be installed as contractually specified.

The manufacturer of the end product is responsible for the correct installation of the engine and the exhaust gas aftertreatment system in the overall system.

The engine and the exhaust gas aftertreatment system may not be modified. In the event of modification, Mercedes-Benz and MTU do not accept responsibility for any damage arising as a result.

Unauthorised conversions or changes to the engine system are not in accordance with regulations, adversely affect safety and may result in the invalidation of EU type approval.

Correct use of the engine system also requires adherence to the instructions in these Operating Instructions. This also requires compliance with the maintenance intervals and the professional execution of maintenance work. Please observe the Workshop Information System (WIS) (▷ page 7).

Implied warranty

! Observe the notes in these operating instructions on correct operation of your assembly as well as on any possible damage to the assemblies. Any damage to your assembly which arises due to culpable breaches of these notes, is not covered by the implied warranty.

Stored data

Several of the electronic components in your engine system contain data memories.

These data memories temporarily or permanently store technical information about:

- the engine system state
- events
- malfunctions

In general, this technical information documents the state of a component, a module, a system or the surroundings.

This includes, for example:

- operating conditions of system components. For example, fluid levels
- the vehicle/device status messages and those of its individual components, e.g. speed, deceleration in movement, accelerator position
- malfunctions and defects in important system components
- the vehicle/device reactions and operating statuses in special driving situations
- ambient conditions, e.g. outside temperature

This data is exclusively technical in nature and can be used to:

- assist in the detection and rectification of faults and defects
- analyse vehicle/device functions, e.g. after an accident

The data cannot be used to trace the vehicle's movements.

When you use one of the available services, technical information may be read from the event data memory and fault data memory.

Services include, for example:

- repair services
- service processes
- implied warranty and guarantee cases
- quality assurance

The information is read out by employees of the service network (including manufacturers) using special diagnostic testers. Further information is available there if required.

After a fault has been rectified, the information is deleted from the fault memory or is continually overwritten.

Qualified specialist workshop

! Have the engine electronics and parts belonging to it such as control units, sensors, actuating components or electric cables maintained only at a qualified specialist workshop. Otherwise, vehicle/device components may wear more quickly and the vehicle's/devices' operating permit may be rendered invalid.

A qualified specialist workshop has the necessary specialist knowledge, tools and qualifications to carry out the work required on the engine to a professional standard. This is particularly applicable to work relevant to safety.

Observe the notes in the Maintenance Booklet.

Always have the following maintenance work carried out at a qualified specialist workshop:

- work relevant to safety
- service and maintenance work
- repair work
- modifications as well as installations and conversions
- work on electronic components

Please consult an MTU or MTU-authorised Mercedes-Benz partner (see addresses in the publication details on the inside of the back cover).

Further applicable documents

These Operating Instructions describe all models and all standard and optional equipment available for your engine system within the scope of delivery of Daimler AG. The installation of the engine system into the vehicle/device may require additional operating instructions adapted to the vehicle/device and the appropriate use thereof. These additional operating instructions will be provided by the vehicle/device manufacturer (manufacturer of the end product). The additional operating instructions will describe, in particular, the functions specific to the installation and operation, the use of such functions as well as warning and control mechanisms.

To use the engine correctly, you also require the Maintenance Booklet.

For US-certified off-highway engines you also require the "Emission Warranty" supplement.

Always keep these documents together with the engine, vehicle or device. These documents should be passed on to the new owner if you sell the engine, vehicle or device.

When carrying out maintenance work, you require access to the Workshop Information System (WIS) via the Internet. This access is subject to a fee.

Current information on the system and prices can be found at this web address: http://service-parts.mercedes-benz.com. Click on "EPC, WIS/ASRA" in the "Service and parts information" tab and then on "WIS".

You can log in by clicking on "Register" on the right-hand side.

General information

The engine is a four-stroke, water-cooled diesel engine with direct injection.

Turbocharging at the engine is handled in a single stage by an exhaust-gas turbocharger.

The engine is equipped with a common rail diesel injection system, cooled and regulated exhaust gas recirculation and exhaust gas turbocharging with charge-air pressure control.

The valve gear has two camshafts at the top which are driven via a gearwheel drive.

The engine features a single-piece cylinder head. In the cylinder head there are two inlet valves and two exhaust valves per cylinder. The valves are arranged symmetrically. The symmetrical valve arrangement is optimal for combustion.

The exhaust gas aftertreatment unit is characterised by the following technologies:

- diesel oxidation catalytic converter (DOC)
- the diesel particle filter (DPF)
- selective catalytic reduction (SCR) with ammonia slip catalytic converter

The engine brake is a decompression brake. It has a controlled exhaust valve which gives it high braking power. The engine brake is controllable in steps or modulated.

The cylinder liners with long-term cylinder socket below in the crankcase lend the engine its compact design.

There is a water separator integrated into the fuel filter housing.

Common rail diesel injection system

The engine features a diesel injection system that uses the Amplified Pressure Common Rail System (APCRS). This minimises the fuel quantity required for combustion. The advantage of this system is that the pressure in the rail and the high-pressure lines is relatively low. The fuel pressure is increased as required in a second stage inside the injector by a pressure transducer. This has a particularly positive effect on material strain and thus on component durability.